



KENYATTA UNIVERSITY
DEPARTMENT OF PHILOSOPHY

UCU 111: INTRODUCTION TO CRITICAL &
PROBLEM SOLVING

By

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TOPIC ONE: PHILOSOPHY

a. What is Philosophy?

1. The word “philosophy” comes from the Greek φιλοσοφία (*philosophia*), which literally means “love of wisdom” The word 'philosophy' is derived from the combination of two ancient Greek

words; '*philos*', which means 'love', and '*sophia*', which means 'wisdom'. It etymologically means love of wisdom.

2. It is a system of beliefs about reality that entails one's integrated view of the world.
3. It is not a body of doctrine, but an activity of actualizing reality that aims at the logical clarification of thoughts.
4. In its broadest and original use, 'philosophy' means the systematic study of the world and our place in it. It entails a critical examination of reality characterized by rational inquiry that aims at the Truth for the sake of attaining wisdom.
5. Is the foundation of knowledge from which all other disciplines derive and the standard by which ideas are integrated and understood

b) Branches of Philosophy

- a. **Epistemology:** The term "epistemology" is derived from the ancient Greek word '*episteme*' meaning 'knowledge. Etymologically it denotes the philosophical study of knowledge and its justification, and of the family of concepts which are involved in our assessing claims to knowledge or justified belief. As a theory of

knowledge, epistemology seeks to establish the process of claiming to know and on what certainty basis are such claims founded. In its broadest sense, epistemology is the study of the method of acquiring and processing knowledge.

b. **Logic:** logic is the study of the methods and principles used to distinguish good (correct) from bad (incorrect) reasoning. In its broadest sense, Logic deals with the study of the evidential link between the premises and conclusions. The study of logic is divided into two; deductive and inductive logic.

c. **Axiology:** (from Greek *axios*, “worthy”; *logos*, “science”) basically refers to the science of values. Various terms have been used in reference to axiology:- Theory Of Value, the philosophical study of goodness, or basically, value. The significance of axiology as a field of study lies first in the considerable expansion of the term that has eventually given a wider meaning to the term value and secondly, in the unification that it has provided for the study of a variety of questions- economic, moral, aesthetic, religious, political and even logical—that had often been considered in relative isolation.

d. **Metaphysics:** is the branch of philosophy responsible for the study of existence. As a term, metaphysics is derived from two ancient Greek words ‘*meta*’ which means beyond and ‘*physicea*’ which refers to material substance or objects of experience. It deals with such abstract conceptual topics/concepts such as *being, existence, immortality, God, spiritual beings, time, identity, consciousness, cause, essence, space, constancy* etc.

USES OF PHILOSOPHY

Much of what is learned in philosophy can be applied in virtually any endeavor. This is both because philosophy touches on so many subjects and especially, because many of its methods are usable in any field.

a. General Uses of Philosophy

- i. *General Problem Solving.* The study of philosophy enhances one's problem solving capacities. It helps one to analyze concepts, definitions, arguments and problems. It contributes to one's capacity to organize ideas and issues, to deal with questions of value and to extract what is essential from masses of information. It helps one both to distinguish fine differences between views and to discover common ground between opposing positions. And it helps one to synthesize a variety of views or perspectives into a unified whole.
- ii. *Communication Skills.* Philosophy contributes to the development of expressive and communicative powers. It provides the basic tools of self-expression - for instance, skills in presenting ideas through well-constructed, systematic arguments. It helps one to express what is distinctive of one's view; enhances one's ability to explain difficult material; and helps one to eliminate ambiguities and vagueness from one's writing and speech.
- iii. *Persuasive Powers.* Philosophy provides training in the construction of clear formulations, good arguments, and apt examples. It helps one develop the ability to be convincing. One learns to build and defend one's own views, to appreciate competing positions, and to indicate forcefully why one considers one's own views

preferable to alternatives. These capacities can be developed not only through reading and writing in philosophy, but also through the philosophical dialogue, in and outside the classroom, that is so much a part of a thoroughgoing philosophical education.

- iv. *Writing Skills.* Philosophy teaches interpretive writing through its examination of challenging texts, comparative writing through emphasis on fairness to alternative positions, argumentative writing through developing students' ability to establish their own views, and descriptive writing through detailed portrayal of concrete examples: the anchors to which generalizations must be tied. Structure and technique, then, are emphasized in philosophical writing. Originality is also encouraged, and students are generally urged to use their imagination and develop their own ideas.

b. Uses of Philosophy in Educational Pursuits

The general uses of philosophy just described are obviously of great academic value. It should also be clear that the study of philosophy has intrinsic rewards as an unlimited quest for understanding of important challenging problems. But philosophy has further uses in deepening an education, both in college and in the many activities, professional and personal, that follow graduation.

- i. *Understanding Other Disciplines.* Philosophy is indispensable for this. Many important questions about a discipline, such as the nature of its concepts and its relation to other disciplines, do not belong to that discipline, are not usually pursued

in it, and are philosophical in nature. Philosophy of science, for instance, is needed to supplement the understanding of the natural and social sciences which one derives from scientific work itself. Philosophy of literature and philosophy of history are of similar value in understanding the arts. Philosophy is moreover, essential in assessing the various standards of evidence used by other disciplines.

Since all fields of knowledge employ reasoning and must set standards of evidence, logic and epistemology have a general bearing on all these fields.

ii. *Development of Sound Methods of Research and Analysis.* Still another value of philosophy in education is its contribution to one's capacity to frame hypotheses, do research, and put problems into manageable form. Philosophical thinking strongly emphasizes clear formulation of ideas and problems, selection of relevant data, and objective methods for assessing ideas and proposals. It also emphasizes development of a sense of the new directions suggested by the hypotheses and questions one encounters in doing research. Philosophers regularly build on both the successes and the failures of their predecessors. A person with philosophical training can readily learn to do the same in any field.

c) Practicality of Philosophy in Non-Academic Careers

i) Philosophy and Job Placement

It should be stressed immediately that the non-academic value of a field of study must not be viewed mainly in terms of its contribution to obtaining one's first job after graduation. Often students are mostly preoccupied with getting their first job,

but even from a narrow vocational point of view it would be short-sighted to concentrate on that at the expense of developing potential for success and advancement once hired. What gets graduates initially hired may not yield promotions or carry them beyond their first position, particularly given how fast the needs of many employers alter with changes in social and economic patterns. It is therefore crucial to see beyond what a job description specifically calls for.

To this end, philosophy need not be mentioned among a job's requirements in order for the benefits derivable from philosophical study to be *appreciated* by the employer, and those benefits need not even be explicitly appreciated in order to be *effective* in helping one advance. It should also be emphasized here that - as recent studies show - employers want and reward many of the capacities which the study of philosophy develops: for instance, the ability to solve problems, to communicate, to organize ideas and issues, to assess pros and cons and to boil down complex data. These capacities represent *transferable skills*. They are transferable not only from philosophy to non-philosophical areas, but from one non-philosophical field to another.

For this reason, people trained in philosophy are not only prepared to do many kinds of tasks; they can also cope with change or even move into new careers more readily than many others.

ii) Philosophy and Business

Regarding current trends in business, a writer in the *New York Times* reported that

"...businessmen are coming to appreciate an education that at its best produces graduates who can write and think clearly and solve problems"
(June 23, 1981).

A recent long-term study by the Bell Telephone Company moreover, determined that majors in liberal arts fields, in which philosophy is a central discipline, "continue to make a strong showing in managerial skills and have experienced considerable business success" (*Career Patterns*, by Robert E. Beck). The study concluded that "there is no need for liberal arts majors to lack confidence in approaching business careers."

A related point was made by a Senior Vice President of the American Can Company:
"Students with any academic background are prepared for business when they can educate themselves and can continue to grow without their teachers, when they have mastered techniques of scholarship and discipline, and when they are challenged to be all they can be" (*Wall Street Journal*, February 2, 1981).

As all this suggests, there are people trained in philosophy in just about every field. They have gone not only into such professions as teaching (at all levels), medicine, and law, but into computer science, management, publishing, sales, criminal justice, public relations, and other fields. Some professionally trained philosophers are also on legislative staffs, and the work of some of them, for a senior congressman, prompted him to say:

c) Principal functions of philosophy

1. ***Integration of experience:*** The universe as we experience it is both diverse and unified. Reasoned thought helps us to integrate and harmonize these apparently opposed aspects of reality.
2. ***Nurturing of our awareness and sensitivity:*** Reasoned thought assists us to not only understand the universe but also ourselves as part of it. Self-examination enhances the consciousness of our own limitations and capabilities. This awareness and sensitivity is crucial in assisting us to adapt to the challenging and complex situations of life.
3. ***Clarification and justification of belief:*** Beliefs are the basis of our actions. Reasoned thought about our beliefs enables us to ensure that they are well-founded and thus rationally justifiable. If our actions are to be effective in enhancing our well-being, they should be founded on explicit, clear, distinct and rationally justifiable beliefs.
4. ***Bridging the gap between theory and practice:*** Philosophy raises questions regarding the meaning, foundations, purpose, justification, verification and application of theories in the pragmatic context. This is useful in ensuring that theories inform practice appropriately. Philosophy also evaluates practice and raises questions regarding the extent to which practice confirms and conforms to theory and why.

5. *Providing a condition for the freedom of the mind:* Philosophy seeks to liberate us from the slavery of ignorance and irrationality. It helps us to examine our own beliefs, assumptions and prejudices. It assists us to act rationally and justly. This in turn broadens our realm of freedom and enhances our capacity to act and respond responsibly, intelligently and creatively

TOPIC TWO: ON ESSENTIALS OF CRITICAL AND CREATIVE THINKING

a. Creative thinking

Creative thinking refers to the formation of possible solutions to a problem or possible explanations of a phenomenon. It entails bringing into being of something which did not exist before, either as a product, a process or a thought, or working on the old with a view of producing a harmonized new.

How is Creativity demonstrated? If we:

- Invent something which has never existed before
- Invent something which exists elsewhere but you are not aware of
- Invent a new process for doing something
- Reapply an existing process or product into a new or different market
- Develop a new way of looking at something (bringing a new idea into existence)
- Change the way someone else looks at something

b. Critical thinking

Whereas Creative thinking is committed to the formation of possible solutions to a problem or possible explanations of a phenomenon, Critical thinking **deals with the testing and evaluation of the proposed solutions.**

Critical thinking varies according to the motivation underlying it. When grounded in selfish motives, it is often manifested in the skillful manipulation of ideas in service to one's own, or one's groups', vested interest. As such, it is typically intellectually flawed, however pragmatically successful it might be. When grounded in fairmindedness and intellectual integrity, it is typically of a higher order intellectually, though subject to the charge of "idealism" by those habituated to its selfish use.

- a. According to Halpern (1996), Critical thinking is the use of those cognitive skills or strategies that increase the probability of a desirable outcome. It is used to describe thinking that is purposeful, reasoned and goal directed - the kind of thinking involved in solving problems, formulating inferences, calculating likelihoods, and making decisions when the thinker is using skills that are thoughtful and effective for the particular context and type of thinking task. Critical thinking also involves evaluating the thinking process - the reasoning that went into the conclusion we've arrived at the kinds of factors considered in

making a decision. Critical thinking is sometimes called directed thinking because it focuses on a desired outcome."

b. To Kurland (1995), Critical thinking is concerned with reason, intellectual honesty, and open-mindedness, as opposed to emotionalism, intellectual laziness, and closed-mindedness. Thus, Critical thinking involves: following evidence where it leads; considering all possibilities; relying on reason rather than emotion; being precise; considering a variety of possible viewpoints and explanations; weighing the effects of motives and biases; being concerned more with finding the truth than with being right; not rejecting unpopular views out of hand; being aware of one's own prejudices and biases, and not allowing them to sway one's judgement."

To this end, we can then define critical thinking as;

.....the intellectually disciplined process of actively and skilfully conceptualising, applying, analysing, synthesizing, and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning, or communication, as a guide to belief and action. In its exemplary form, it is based on universal intellectual values that transcend subject matter divisions: clarity, accuracy, precision, consistency, relevance, sound evidence, good reasons, depth, breadth, and fairness. It entails the examination of those structures or elements of thought implicit in all reasoning: purpose, problem, or question-at-issue, assumptions, concepts, empirical

grounding; reasoning leading to conclusions, implications and consequences, objections from alternative viewpoints, and frame of reference.

Critical thinking - in being responsive to variable subject matter, issues, and purposes - is incorporated in a family of interwoven modes of thinking, among them: scientific thinking, mathematical thinking, historical thinking, anthropological thinking, economic thinking, moral thinking, and philosophical thinking.

As a way of thinking, Critical thinking has two components:

1. a set of skills to process and generate information and beliefs, and
2. the habit, based on intellectual commitment, of using those skills to guide behaviour.

c. The main characteristics of Critical and Creative thinking (Dimensions of Critical thinking)

Wade (1995) identifies 8 (Eight) characteristics of Critical thinking. Critical thinking involves asking questions, defining a problem, examining evidence, analyzing assumptions and biases, avoiding emotional reasoning, avoiding oversimplification, considering other interpretations, and tolerating ambiguity. Dealing with ambiguity is also seen by Strohm & Baukus (1995) as an essential part of Critical thinking, "Ambiguity and doubt serve a critical-thinking function and are a necessary and even a productive part of the process" (p. 56).

Another characteristic of Critical thinking identified by many sources is metacognition. Metacognition is thinking about one's own thinking. More specifically, "metacognition is being aware of one's thinking as one performs specific tasks and then using this awareness to control what one is doing" (Jones & Ratcliff, 1993, p. 10).

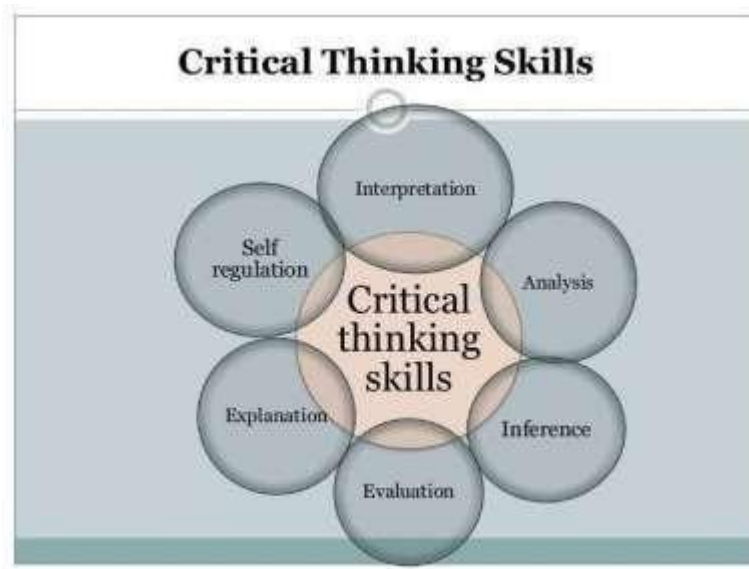
d. Essential aspects of Critical thinking

In the book, *Critical thinking*, Beyer elaborately explains what he sees as essential aspects of Critical thinking. These are:

- **Dispositions:** Critical thinkers are sceptical, open-minded, value fair-mindedness, respect evidence and reasoning, respect clarity and precision, look at different points of view, and will change positions when reason leads them to do so.
- **Criteria:** To think critically, must apply criteria. Need to have conditions that must be met for something to be judged as believable. Although the argument can be made that each subject area has different criteria, some standards apply to all subjects. "... an assertion must... be based on relevant, accurate facts; based on credible sources; precise; unbiased; free from logical fallacies; logically consistent; and strongly reasoned" (p. 12).
- **Argument:** Is a statement or proposition with supporting evidence. Critical thinking involves identifying, evaluating, and constructing arguments.
- **Reasoning:** The ability to infer a conclusion from one or multiple premises. To do so requires examining logical relationships among statements or data.

- **Point of View:** The way one views the world, which shapes one's construction of meaning. In a search for understanding, critical thinkers view phenomena from many different points of view.
- **Procedures for Applying Criteria:** Other types of thinking use a general procedure. Critical thinking makes use of many procedures. These procedures include asking questions, making judgements, and identifying assumptions.

e. Critical thinking Skills



f. The value and importance of Critical and Creative Thinking

1. Critical and Creative Thinking inculcates habits of thought- the ability to think outside the box and use reason as the locomotive to existence.

2. Encourages breadth of vision- the ability to see things from different points of view
3. It gives us a sense of self-perspective in our activities, and our lives: the ability to examine alternatives and choose those relevant to ensure we live an examined and worth life.
4. Its concerns are very abstract but with immense practical value: the ability to translate ideas (theory) into practice through action.
5. Its interest and value lies in its ability in helping us to understand ourselves and our world better and more deeply than we otherwise would, and in permanently altering our approach to our lives and our relations to others through encouraging a lifelong habit of reflection on them.
6. The study of Critical and Creative Thinking is a process whose continuance can immensely enrich one's life, and can open to one view that would otherwise be closed or overlooked.
7. Its serious study can greatly enhance one's analytical, critical, evaluative and interpretive abilities,
8. Its grasp can greatly enhance one's ability to express oneself clearly and to formulate and respond to arguments in speech and writing effectively.
9. Critical and Creative Thinking provides one with general problem-solving skills, skills in analysing concepts, definitions, arguments and problems.
10. It enables one to organize ideas and issues and to extract what is central to an issue from a mass of information.

11.It enhances one's ability to question deeply one's own framework of thought

12.Critical and creative thinking equips an individual with the ability to reconstruct sympathetically and imaginatively the strongest versions of points of view and frameworks of thought opposed to one's own

13.It inculcates in one the ability to reason dialectically (multi-logically) in such a way as to determine when one's own point of view is at its weakest and when an opposing point of view is at its strongest

14.It helps one both to make fine distinctions and to find what common ground between opposing positions is.

15.Critical and Creative thinking encourages one to synthesize or bring together a range of different views into one more comprehensive and coherent position.

16.Critical and Creative Thinking improves one's communication skills, through improving one's ability to present ideas in well-constructed, systematic arguments, to express what is unique about one's views, and to explain difficult material.

17.These skills in presenting well-thought-out arguments, clear formulations, and apt examples, in turn lend one's arguments persuasive power.

18.The give and take of critical and creative thinking based discussions improves one's ability to think on one's feet, and to indicate why one's own views are to be preferred to others.

19. Critical and creative thinking aids one in recognizing when and in what respect one's own views may be incorrect, and what must be revised or discarded and what can be retained.
20. Critical and Creative Thinking lays emphasis on clarity and rigor of argument, the apt use of example and illustration, and sensitivity to the strengths and weaknesses both of views one is examining and of one's own view thus making ideas that determine good living more explicit and relevant.
21. Critical and creative thinking plays an important role in social change-institutions in any society (courts, governments, schools, businesses etc) - are products of a certain way of thinking.
22. Critical and creative thinking helps us in uncovering biases and prejudices.
23. Critical and creative thinking is a path to freedom from half-truths and deceptions as it opens up one's mind in seeing things independent of emotions, native biases and from different perspectives.
24. The willingness to change one point of view as we continue to examine and re-examine ideas that may seem obvious is a major component of critical and creative thinking. Such thinking takes time and the willingness to say three subversive words: "*I don't know.*" Means need to learn and not an expression of ignorance.

6. Essential characteristics of a critical and creative thinker

1. believes in supremacy of reason as the beacon of existence
2. willingness to spend time reflecting on the ideas presented

3. ability to evaluate and solve problems as they come
4. logical and sequential thinking without appealing to emotions
5. diligent in seeking out the truth
6. eager to express their thoughts on a topic
7. exercise the highest level of patience
8. ability to tolerate ambiguity
9. seekers of alternative views on a topic
10. ability to persevere and endure with a view of attaining truth
11. intellectual courage as to pursue issues to their conclusion end
12. open to new ideas that may not necessarily agree with their previous thought on a topic
13. ability to base their judgements on ideas and evidence
14. ability to recognize errors in thought and persuasion
15. ability to recognize good arguments from bad and fallacious arguments
16. willingness to take a critical stance on issues
17. willingness to accept mistakes, correct them and apologise
18. ability to ask penetrating and thought-provoking questions to evaluate ideas
19. ability to being in touch with their personal thoughts and ideas about a topic
20. willingness to reassess their views when new or discordant evidence is introduced and evaluated
21. ability to identify arguments and issues

22.ability to see connections between topics and use knowledge from other disciplines to enhance their reading and learning experiences

TOPIC THREE: SOCRATES

Socrates: *On Socratic Teaching Method*

The oldest and still the most powerful, teaching tactic for fostering Critical and Creative thinking is Socratic teaching. Socratic teaching focuses on giving students questions, not answers; it models an inquiring, probing mind by continually probing into the subject with pertinent questions aimed at nurturing and cultivating the mind into self-knowing. It's to this background that Socrates considered himself *the midwife of knowledge* contrary to what early sophists had conceived a teacher and dispenser of knowledge to be (a custodian that injects and indoctrinates others with what he/she knows or considers to be or to constitute knowledge). The abilities we gain by focusing on the elements of reasoning in a disciplined and self- assessing way, and the logical relationships that result from such disciplined thought, prepare us for Socratic questioning/teaching method that characterise the subject of critical and creative thinking.

Socrates discovered that by a method of systematic probing questionings, people could not rationally justify their confident claims to knowledge. Confused meanings, inadequate evidence, or self-contradictory beliefs often lurked beneath smooth but largely empty rhetoric. Socrates established the fact that one cannot depend upon those in "authority" to have sound knowledge and insight. He demonstrated that persons may have power and high position and yet be deeply confused and irrational.

He established the importance of asking deep questions that probe profoundly into thinking before we accept ideas as worthy of belief. He

established the importance of seeking evidence, closely examining reasoning and assumptions, analyzing basic concepts, and tracing out implications not only of what is said but of what is done as well. His method of questioning is now known as "Socratic questioning" and is the best known Critical and Creative thinking teaching strategy. In his mode of questioning, Socrates highlighted the need in thinking for clarity, precision and logical consistency.

Socrates set the agenda for the tradition of Critical and Creative thinking, namely, to reflectively question common beliefs and explanations, carefully distinguishing those beliefs that are reasonable and logical from those which--however appealing they may be to our native egocentrism, however much they serve our vested interests, however

As a way of Socratic teaching for thinking, critical thinking has two components:

3. a set of skills to process and generate information and beliefs, and
4. the habit, based on intellectual commitment, of using those skills to guide behaviour.

Thankfully, there is a predictable set of relationships that hold for all subjects and disciplines. This is given in the general logic of reasoning, since every subject has been developed by those who had:

- shared goals and objectives (which defined the subject focus)
- shared questions and problems (whose solution they pursued)

- shared information and data (which they used as an empirical basis)
- shared modes of interpreting or judging that information
- shared specialized concepts and ideas (which they used to help them organize their data)
- shared key assumptions (that gave them a basis from which to collectively begin)
- a shared point of view (which enabled them to pursue common goals from a common framework)

A Socratic questioner should:

- a) keep the discussion focused
- b) keep the discussion intellectually responsible
- c) stimulate the discussion with probing questions
- d) periodically summarize what has and what has not been dealt with and/or resolved
- e) draw as many students as possible into the discussion.

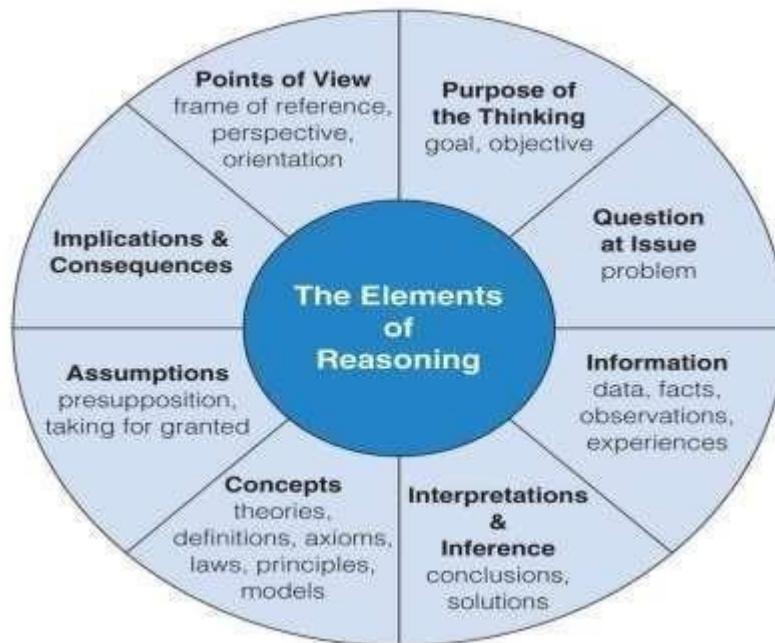
TOPIC FOUR: TOOLS OF CRITICAL AND CRATIVE THINKING

a. Tools of Analysis (Elements of Critical Thinking)

Critical and creative thinking tools of analysis are also referred to as Dimensions of thinking. They entail those essential components that must be taken into consideration when confronted with issues that begs for an answer. The tools of analysis deals with the informational basis to which we can proceed in determining a solution based on inclusive and conclusive knowledge. They include:

- Purpose
- Issue, problem or Question at hand
- Assumptions
- Points of view
- Data, Information and evidence base
- Concepts and ideas
- Inferences or interpretations
- Implications and Consequences

Table: Elements of Thinking



ii) Tools of Evaluation

Critical and creative thinking tools of evaluation are also referred to as Universal intellectual standards. These are standards which must be applied to thinking whenever one is interested in checking the quality of reasoning about a problem, issue, or situation. To think critically entails having command of these standards, being able to make sound decisions, being able to pose questions which probe our thinking- questions which hold us accountable for our thinking, questions which, through disciplined consistency becomes internalised as questions we need to ask ourselves.

The ultimate goal, then, is for these questions to become infused in our thinking, forming part of our inner voice, which then guides us to better reasoning. While there are a number of universal standards, the following are the most significant:

- a. **Clarity:** Could you elaborate further on that point? Could you express that point in another way? Could you give me an illustration? Could you give me an example? Clarity is the gateway standard. If a statement is unclear, we cannot determine whether it is accurate or relevant. In fact, we cannot tell anything about it because we don't yet know what it is saying. For example, the question, "What is thinking?" seems to be unclear. In order to address the question adequately, however, we would need to have a clearer understanding of what the person asking the question is considering the "problem" to be. A clearer question might be "What can the teachers of critical and creative thinking do to ensure that the process both succeeds and answers to the needs of different disciplines and across the curriculum?"
- b. **Accuracy:** Is that really true? How could we check that? How could we find out if that is true? A statement can be clear but not accurate, as in "You are either with us or against us."
- c. **Precision:** Could you give more details? Could you be more specific?- A statement can be both clear and accurate, but not precise, as in "Saddam supports terrorism." (We don't know how he does this; training, arming etc.)
- d. **Relevance:** How is that connected to the question? How does that bear on the issue? - A statement can be clear, accurate, and precise, but not relevant to the question at issue. For example, most globalising agents often think that the amount of effort they put into the economic arm of globalisation should be

used in emphasizing the worth of the process since it opens up international markets and improves the economic standards of the recipient nations. Often, however, the "effort" does not measure the quality of life of the people, and *when this is so*, effort is irrelevant to their appropriate well-being.

- e. **Depth:** How does your answer address the complexities in the question? How are you taking into account the problems in the question? Is that dealing with the most significant factors? - A statement can be clear, accurate, precise, and relevant, but superficial (that is, lack depth). For example, the statement "Just say No" which is often used to discourage children and teens from using drugs, is clear, accurate, precise, and relevant. Nevertheless, it lacks depth because it treats an extremely complex issue, the pervasive problem of drug use among young people, superficially. It fails to deal with the complexities of the issue.
- f. **Breadth:** Do we need to consider another point of view? Is there another way to look at this question? What would this look like from a conservative standpoint? What would this look like from the point of view of...? - A line of reasoning may be clear, accurate, precise, relevant, and deep, but lack breadth (as in an argument from either the conservative or liberal standpoint which gets deeply into an issue, but only recognizes the insights of one side of the question.)
- g. **Logic:** Does this really make sense? Does that follow from what you said? How does that follow? But before you implied this and now you are saying that; how can both be true?

iii) Tools of Transformation

Critical and creative thinking tools of Transformation are also referred to as Valuable intellectual traits. These are transformational traits exhibited by those individuals that embrace criticality and creativity. They are virtue traits that define maturity in reason, objectivity in thought, rationality in deliberations and sobriety save to mention consistency.

- ***Intellectual Humility:*** Having the knowledge of the limits of one's knowledge, including sensitivity to circumstances in which one's native egocentrism is likely to function self-deceptively; sensitivity to bias, prejudice and limitations of one's viewpoint. Intellectual humility depends on recognizing that one should not claim more than one actually knows. It does not imply spinelessness or submissiveness. It implies the lack of intellectual pretentiousness, boastfulness, or conceit, combined with insight into the logical foundations, or lack of such foundations, of one's beliefs.
- ***Intellectual Courage:*** Having the knowledge of the need to face and fairly address ideas, beliefs or viewpoints toward which we have strong negative emotions and to which we have not given a serious hearing. This courage is connected with the recognition that ideas considered dangerous or absurd are sometimes rationally justified (in whole or in part) and that conclusions and beliefs inculcated in us are sometimes false or misleading. To determine for ourselves which is which, we must not passively and uncritically "accept" what we have "learned." Intellectual courage comes into play here, because inevitably we will come to see some truth in some ideas considered dangerous

and absurd, and distortion or falsity in some ideas strongly held in our social group. We need courage to be true to our own thinking in such circumstances. The penalties for non-conformity can be severe.

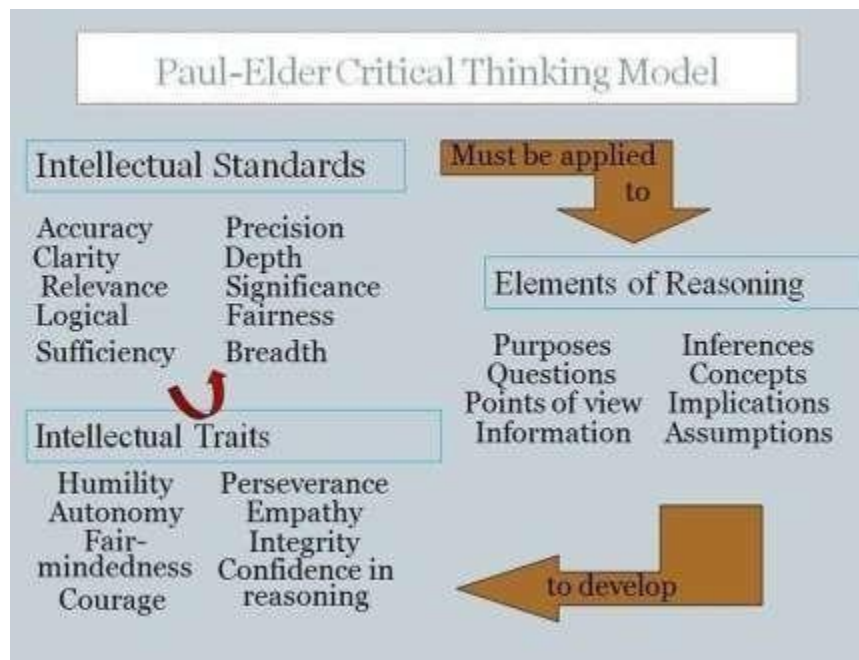
- ***Intellectual Empathy:*** Having the knowledge of the need to imaginatively put oneself in the place of others in order to genuinely understand them, this requires the consciousness of our egocentric tendency to identify truth with our immediate perceptions of long-standing thought or belief. This trait correlates with the ability to reconstruct accurately the viewpoints and reasoning of others and to reason from premises, assumptions, and ideas other than our own. This trait also correlates with the willingness to remember occasions when we were wrong in the past despite an intense conviction that we were right, and with the ability to imagine our being similarly deceived in a case-at-hand.
- ***Intellectual Integrity:*** Recognition of the need to be true to one's own thinking; to be consistent in the intellectual standards one applies; to hold one's self to the same rigorous standards of evidence and proof to which one holds one's antagonists; to practice what one advocates for others; and to honestly admit discrepancies and inconsistencies in one's own thought and action.
- ***Intellectual Perseverance:*** Having the knowledge of the need to use intellectual insights and truths in spite of difficulties, obstacles, and

frustrations; firm adherence to rational principles despite the irrational opposition of others; a sense of the need to struggle with confusion and

unsettled questions over an extended period of time to achieve deeper understanding or insight.

- ***Faith In Reason:*** Confidence that, in the long run, one's own higher interests and those of humankind at large will be best served by giving the freest play to reason, by encouraging people to come to their own conclusions by developing their own rational faculties; faith that, with proper encouragement and cultivation, people can learn to think for themselves, to form rational viewpoints, draw reasonable conclusions, think coherently and logically, persuade each other by reason and become reasonable persons, despite the deep-seated obstacles in the native character of the human mind and in society as we know it.
- ***Fairmindedness:*** Having the knowledge of the need to treat all viewpoints alike, without reference to one's own feelings or vested interests, or the feelings or vested interests of one's friends, community or nation; implies adherence to intellectual standards without reference to one's own advantage or the advantage of one's group.

Model for Critical Thinking



Adopted from Paul and Elder 1998

TOPIC FIVE: EDUCATION, IDENTITY AND SELF EXAMINATION

a. Definition of Education Education

- Education is the process through which knowledge, skills and information is transmitted from one person to the other.
- It is the knowledge acquired by an individual after studying particular subject matter or experiencing life lessons that provide understanding of oneself, environment and the world view.
- It is a state in which a person's state of mind is optimised regardless of the situation as to be able to perceive accurately, think clearly and act effectively in order to achieve self-selected goals and aspirations

- d) Is the process of growth aimed at nurturing mental abilities and acquisition of skills requisite for rational existence.

b. Role of Education

Education;

- a. liberates an individual from ignorance and dogmatic tendencies to rationality and self-understanding
- b. Transforms an individual from a state of individualism and mass-manship to autonomy and individuality
- c. Integrates an individual with himself, nature and society
- d. is a catalyst to progress and positive change for both the individual and society
- e. is a catalyst to development and reengineering of human society and individual growth
- f. is a means to self-actualization, self-discovery and reinvention
- g. is a lever for human emancipation from the fetters of self mis-definition, self-misconception and self-deception.

Essential Skills for Thinking

There are a wide range of skills, dispositions and attitudes which together characterize a virtue which has both intellectual and moral aspects, and which serves to prevent the emergence of numerous vices, including dogmatism and prejudice. One central purpose of education is to prepare students to be able to form "a reasonable judgement on controversial questions in regard to which they are likely

to have to act". In addition to having access to impartial supplies of knowledge, education needs to offer "training in judicial habits of thought. Beyond access to such knowledge, students need to develop certain *skills* if the knowledge acquired is not to produce individuals who passively accept the teacher's wisdom or the creed which is dominant in their own society; let's talk of intelligence, contrasting it with information alone; to indicate the whole set of critical abilities such education is supposed to impact.

Such critical skills grounded in knowledge include:

1. *the ability to form an opinion for oneself*, which involves being able to recognize what is intended to mislead, being capable of listening to eloquence without being carried away, and becoming adept at asking and determining if there is any reason to think that our beliefs are true;
2. *the ability to find an impartial solution*, which involves learning to recognize and control our own biases, coming to view our own beliefs with the same detachment with which we view the beliefs of others,

judging issues on their merits, trying to ascertain the relevant facts, and the power of weighing arguments;

3. *the ability to identify and question assumptions*, which involves learning not to be credulous, applying constructive doubt in order to test unexamined beliefs, and resisting the notion that some authority, a great philosopher perhaps, has captured the whole truth.

Dispositions for critical Thought

The mere possession of critical skills is not sufficient to make one a critical thinker or full developed individual. We must draw our attention to certain *dispositions* which will guarantee that the relevant skills are actually exercised. Many names can be used to refer to these disposition- *habits* or the notion of *practice* to suggest the translation of skills into actual behaviour. In this respect education can be understood as the formation, by means of instruction, of certain mental habits [and a certain outlook on life and the world]. Of great importance among these dispositions are:

- i. *the habit of impartial inquiry*, which is necessary if one-sided opinions are not to be taken at face value, and if people are to arrive at conclusions which do not depend solely on time and place of their education;
- ii. *the habit of weighing evidence*, coupled with the practice of not giving full assent to propositions which there is no reason to believe true;
- iii.(iii) *the habit of attempting to see things truly*, which contrasts with the practice of merely collecting whatever reinforces existing prejudice; and
- iv.(iv) *the habit of living from one's own centre*, a kind of self-direction, a certain independence in the will. Such habits, of course, have to be exercised intelligently. This can however be problematic and it's the duty of Critical thinking to address. One becomes a victim of habit if the habitual beliefs of one's own age constitute a prison of prejudice. Hence the need for a *critical* habit of mind.

Because they are not simply automatic responses in which one has been drilled, such intellectual habits in effect reflect a person's willingness; call it one's *readiness*, to act and respond in various ways. And examples here could include:

(i) *a readiness to admit new evidence against previous beliefs*, which involves an open-minded acceptance (avoiding credulity) of whatever a critical examination has revealed;

(ii) *a readiness to discard hypotheses which have proved inadequate*, where the test is whether or not one is prepared in fact to abandon beliefs which once seemed promising; and

(iii) *a readiness to adapt oneself to the facts of the world*, as opposed to merely going along with whatever happens to be in the ascendant, which might be evil. To be ready to act, or react in this way suggests both an awareness that the habits in question are appropriate and a principled commitment to their execution is justifiable. These dispositions have in common the virtue of *truthfulness*, which entails the wish to find out, and trying to be right in matters of belief.

c. Traits of a Critical Mind

Beyond the skills and dispositions outlined above, a certain set of *attitudes* characterize the outlook of a critical person. By critical attitude we mean a temper of mind central to which a certain stance with respect to knowledge and opinion; it involves:

- (i) *a realization of human fallibility*, a sense of the uncertainty of many things commonly regarded as indubitable, bringing with it humility;
- (ii) *an open-minded outlook* with respect to our beliefs, an "inward readiness" to give weight to the other side, where every question is regarded as open and where it is recognized that what passes for knowledge is sure to require correction;
- (iii) *a refusal to think that our own desires and wishes provide a key to understanding the world*, recognizing that what we should like has no bearing whatever on what is;
- (iv) *being tentative, without falling into a lazy scepticism* (or dogmatic doubt), but holding one's beliefs with the degree of conviction warranted by the evidence. This entails having a strong desire to know combined with great caution in believing that one knows must meet the test of certainty and proof. It entails assurance that open-mindedness does not become regenerate into being mindless.

TOPIC SIX: FALLACIES

a. **Definition:** Fallacies are errors in reasoning which from the surface value looks accurate and convincing but when exposed to retrospection we realise that reason has been flawed.

b. Division of Fallacies

Fallacies are generally divided into two broad groups - formal and informal.

- d. Formal fallacies are most conveniently discussed in connection with certain patterns of valid inference to which they bear a superficial resemblance.

- e. Informal fallacies on the other hand, are errors in reasoning into which we may fall either because of carelessness and inattention to our subject matter or through being misled by some ambiguity in the language used to formulate our arguments.

Informal fallacies are divisible into two - **fallacies of relevance** and **fallacies of ambiguity**.

a. Fallacies of Relevance

Common to all arguments that commit fallacies of relevance is the fact that their premises are logically irrelevant to, and therefore incapable of establishing the truth of their conclusions. The irrelevance here is logical rather than psychological, of course, for unless there are some psychological connections, there would be no persuasiveness or seeming correctness. How psychological relevance can be confused with logical relevance is explained in some cases by reference to the fact that language can be used expressively as well as informatively, to stimulate such emotions as fear, hostility, pity, enthusiasm or awe.

- ***Argumentum ad Baculum (appeal to force)***: This is a fallacy committed when one appeals to force or the threat of force to cause acceptance of a conclusion. It is usually resorted to only when evidence or rational arguments fail.
- ***Argumentum ad ignorantiam (argument from ignorance)***: This fallacy is illustrated by the argument that there must be ghosts because no one has ever been able to prove that there aren't any. It is committed whenever it is argued that a proposition is true simply on the basis that it has not been proved false, or that it is false because it has not been proved true.

- ***Argumentum ad misericordiam (appeal to pity)***: This is a fallacy committed when pity is appealed to for the sake of getting a conclusion accepted. This argument is frequently encountered in courts of law, when a defence attorney may disregard the facts of the case and seek to win his client's acquittal by arousing pity in the jurymen.
 - ***Argumentum ad populum (appeal to the masses)***: Sometimes it is defined as the fallacy committed in directing an emotional appeal "to the people" or "to the gallery" to win their assent to a conclusion unsupported by good evidence. That is, an attempt to win popular assent to a conclusion by arousing the feelings and enthusiasms of the multitude. This is favourite device with the propagandists, politicians, street preachers, demagogues, and the advertisers.
 - ***Argumentum ad verecundiam (appeal to authority)***: This is the appeal to authority- that is, to the feeling of respect people have for the famous-to win assent to a conclusion. This method of argument is not always strictly fallacious, for the reference to an admitted authority in the special field of his competence may carry great weight and constitute relevant evidence. etc
- b. **Fallacies of ambiguity**: "fallacies of ambiguity" or "fallacies of clearness." occur in arguments whose formulations contain ambiguous words or phrases, whose meanings shift and change more or less subtly in the course of the argument and thus renders it fallacious. These include:

- ***Fallacy of Equivocation:*** Most words have more than one literal meaning, as the word “pen” which may denote either an instrument for writing or an enclosure for animals. When we keep these two meanings apart, no difficulty arises. But when we confuse the different meanings a single word or phrase may have, using it in different senses in the same context, we are using it equivocally.

A traditional example of this fallacy is:

The end of a thing is its perfection

Death is the end of life

Therefore, death is the perfection of life

This argument is fallacious because two different senses of the word “end” are confused in it. The word end may mean either “goal” or “last event.” Both meanings are of course legitimate. But what is illegitimate is to confuse the two, as

in this argument. The premises are plausible only when the word “end” is interpreted differently in each of them, as: “the goal of a thing is its perfection,” and “death is the last event of life.”

Fallacy of Amphiboly: This fallacy occurs in arguing from premises whose formulations are ambiguous because of their grammatical construction. A statement is amphibolous when its meaning is indeterminate because of the loose or awkward way in which its words are combined. An amphibolous statement may be true on one interpretation and false on another. When it is stated as the premise with the interpretation which makes it true, and a conclusion is drawn from it on the

interpretation which makes it false, then the fallacy of amphiboly has been committed.

- ***Fallacy of Accent:*** This is committed in an argument whose deceptive but invalid nature depends upon a change or a shift in meaning. The way in which meaning shifts depends upon what part of it may be emphasised or accented. Some statements have quite different meanings when different words are stressed.
- ***Fallacy of Composition:*** Fallacy of composition is applied to two closely related types of invalid argument. The first may be described as reasoning fallaciously from the properties of the parts of the whole to the properties of the whole itself. For example to argue that since every part of a particular machine is light in weight, the machine as a whole is light in weight.

c. How to avoid Fallacious Thinking

Fallacies are pitfalls into which any of us may tumble in our reasoning. There is no sure way to avoid fallacies. To avoid fallacies of relevance requires constant vigilance and awareness of the many ways in which irrelevance can intrude. A realization of the flexibility of language and the multiplicity of its use will keep us from mistaking an exhortation to accept and approve a conclusion for an argument designed to prove that conclusion true.

The fallacies of ambiguity are subtle things. Words are slippery, and most of them have a variety of different senses or meanings. Where these different meanings are confused in the formulation of an argument, the reasoning is fallacious. To avoid the various fallacies of ambiguity, we must have and keep the meanings of our terms clearly in mind. One way to accomplish this is by defining the key terms that are used. Since shifts in the meanings of terms can make arguments fallacious, and since ambiguity can be avoided by careful definition of the terms involved, definition is an important matter for anyone interested in the study of logic.

TOPIC SEVEN: APPEARANCE AND REALITY - Bertrand Russell

- Appearance (that which we perceive, our sensations of the empirical objects, that which is experienced by our senses)
- Reality (the actuality of beingness, the thing in itself, the essence of reality, comprehended by the mind).
- **The problem of appearance-**
 - a. That senses are at times deceptive.
 - b. It is prudent for man not to trust where he has once been deceived.
 - c. The mind replicates the objects of nature
 - d. Mistaken identities are common mental phenomena.
 - e. An unnurtured conscience is subject to error
 - f. Knowledge is dynamic; it grows and changes with new occurrences.

- g. There are a myriad other impediments to clear thinking; authority, paternalism, religion etc.
- h. Propaganda mistaken for knowledge- by conmen, politicians, propagandists etc

THUS: We incessantly misconstrue Appearance for Reality.

. Importance of distinguishing Appearance from Reality

- a. Better understanding of ourselves and the world we inhabit
- b. More knowledgeable about the essence of being and the nature of things
- c. Catalyst to a well defined form of life.
- d. Avoidance of errors characteristic of mistaken identities.
- e. It gives meaning and relevance to the rational nature of man.

TEXT REFERENCE: Is there any knowledge in the world which is so certain that no reasonable man could doubt it?

In daily life, we assume as certain many things which, on a closer scrutiny, are found to be so full of apparent contradictions that only a great amount of thought enables us to know what it is that we may believe. In the search for certainty, it is natural to begin with our present experiences, and in some sense; no doubt, knowledge is to be derived from them.

TOPIC EIGHT: JOHN STUART MILL: ON LIBERTY OF THOUGHT AND INDIVIDUALITY AS ELEMENTS OF HUMAN WELL-BEING

A. J. S. Mill's opening Statement in *On Liberty of Thought and Individuality*

“The time, it is to be hoped, is gone by when any defence would be necessary of the “liberty of the press” as one of the securities against corrupt or tyrannical government. No argument, we may suppose, can now be needed against permitting a legislature or executive, not identified in interest with people, to prescribe opinions to them and determine what doctrines or what arguments they shall be allowed to hear. This aspect of the question, besides, has been so often and so triumphantly enforced by preceding writers that it need not be specially insisted on this place. Speaking generally, it is not, in constitutional countries, to be apprehended that the government, whether completely responsible to the people or not, will often attempt to control the expression of opinion, except when in doing so it makes itself the organ of the general intolerance of the public. Let us suppose, therefore, that the government is entirely at one with the people, and never thinks of exerting any power of coercion unless in agreement with what

it conceives to be their voice.” (J.S. Mill’s On Liberty of Thought)

B. key words:

- a. **Freedom:** The liberties enjoyed by a person, it entails the liberties to hold opinion, express oneself, association, choice, etc
- b. **Definitions of Individuality and autonomy** - thinking for oneself, ability to stand out, creativity, productiveness, independence of thought, distinctive innovativeness etc. It also entails being considerate of others and contributing to the wellbeing of others independent of external influence or forces.
- c. **Mills’ purpose:** to discuss the degree to which the government and society may interfere in the lives of the citizens. The argument is that as human beings we are endowed with the ability to think and determine the destiny of our own lives:- that any interference by the government in the freedom and liberties of individual persons is only necessary if it aims at preventing one person from harming the other but such interference cannot be warranted where the government has as its aim the appropriation of the freedom and liberties of individual persons and gag their creative potentials.
- d. That our freedom and liberties are our entitlement only insofar as they do not infringe on the liberties and freedoms of other persons.

- e. That as much as we may enjoy the liberties and freedom to do as we wish and act the way we act, our actions are enshrined in our liberties if they do not affect, in the negative way the interests and liberties of others.
- f. Any action we undertake in the name of our freedoms and liberties is therefore naturally restricted by way of the impact it exerts on the freedoms and liberties of others.
- g. Freedom is only freedom insofar as it favours the exercise of freedom and liberties by others, without causing inconveniences in the process of enjoying and executing it.

The maxim: Do to others what you would want others to do unto you.

Summary of *On Liberty*

In *On Liberty* Mill discusses the degree to which the government and society may interfere in the lives of citizens. He argues that such interference is warranted only to prevent one person from harming another. Compelling someone to act for his own good, or to prevent him from harming himself, is never justified. (*Berger, 1984*). According to Mill people should be allowed to think and speak, as they like, to choose their own way of living, and to choose their associates. And because Mill's principle draws the line at harm to others- a pure utilitarian principle (utilitarianism

is the theory that we should strive to produce as much happiness as possible), it is often called the “harm principle.”

TOPIC NINE: BARRIERS TO CRITICAL AND CREATIVE THINKING

Optimistic assessments of human curiosity and despairing visions of stone-faced ordinary humans and college students are genuinely true. Human beings are born with an innate curiosity and sense of wonder, but these must be nurtured if they are to survive.

Ever since the ancient times, Humans have been considered to be rational beings. Rational in the sense, that their actions are propelled by reason. The very fact that human beings punctuate their actions with reason is a fundamental distinguishing element between man and the rest of earthly species. But despite the belief that man is rational, most of his actions cast aspersions on the true nature of his rational component. Barnes', for example (1992), rightly observes that Man is the only animal that uses meanings- ideas, concepts, analogies, metaphors, models, theories, and explanations- to make sense of things and to understand, predict, and control things. He is also the only creature that uses meanings to negate, contradict, and deceive themselves, to misconceive, distort, and stereotype, and to be dogmatic, prejudiced, and narrow-minded. Humans are also the only animals whose thinking can be characterized as clear, precise, accurate, relevant, consistent, profound, and fair; at the same time, they are the only animals whose thinking is often imprecise, vague, inaccurate, irrelevant, superficial, trivial, and biased. Man is a thinking being

- The damage done by multiple forms of prejudice and narrow-mindedness- academic, social, personal, professional, religious, racial, national, and ideological- continues to mount.
- Culture and traditions
- Religious fanaticism and dogmas
- Paternalism
- School environment
- Peer and social influences
- Authoritarian regimes and/or systems
- Negative preoccupations in our lives – wrong choices
- Propaganda: the deliberate, systematic attempt to shape perceptions, manipulate cognitions, and direct behavior to achieve a response that furthers the desired intent of the propagandist
- Inherent and perpetual fear of the unknown
- Mental laziness/ perpetual dogmatic slumber

TOPIC TEN: DECISION MAKING AND PROBLEM SOLVING

Definitions

i. Problem

The word *problem* is used in the broad sense: one has a problem when one has a need or question but no obvious answer to it. In this case, all mental insatisfactions and the quest to grasp the essence of the unknown; be it physical or psychological fall within domain of what is rightly defined as “*problem*”.

ii. Decision Making

Decision making can be defined as the mental processes (cognitive process) resulting in the selection of a course of action among several alternative scenarios. Every decision making process produces a definite solution that characterize our actions, opinions and choice.

Phases Of Decision Making And Problem Solving

The general procedure for applying critical thinking to any problem can be described as a cycle with five phases. This cycle should however not be treated as a rigid procedure in which each phase must be complete before the next is begun. In practice, you may have to go back to the earlier phase or work on several phases simultaneously. But if you need to have any real assurance that your ultimate

decision is sound, then all phases must be complete. The details of each phase may vary depending on the problem at stake, but the general principles apply to all situations.

1. The first phase of problem solving involves *recognition and definition of the issue at stake*. Generally speaking, a typical process of decision-making begins with the recognition of a problem. It is commonly true, that many problems are never solved because they are not recognized soon enough or not recognized at all. For example, some freshmen fail in college because they do not recognize soon enough that their study habits are inadequate or that they are in an unsuitable curriculum.

Once a problem has been recognized, it should be carefully defined. Failure to attain a clear definition of a problem will always result in obtaining unsuccessful solutions or you may end up solving “some” problem but not the one that you were trying to solve.

In many situations, defining the problem will be the most difficult phase in decision-making. But once you have correctly defined the problem, the rest will be relatively easy.

There are three rules that must be followed in defining the problem.

- i. *the definition should not be too general*. This is true because if the definition is too broad, the guidelines for a solution will be too broad, and the investigation may flounder. Large problems can be very real, but their solution usually requires

breaking them down into smaller, clearly defined segments in order to solve them one at a time.

- ii. *the definition should not be too specific.* A definition of a problem is said to be too specific when it unnecessarily restricts alternative solutions. When the definition of the problem is too specific, it will always lead to temporary solutions because it will have ignored other significant aspects that led to its emergence.
- iii. *the definition should not in itself constitute a “solution” to the problem.* Suppose that in each year, there is a problem of mass drop-out of Masters Students in the School of Humanities and Social Sciences at Kenyatta university in Kenya, and the Dean of School defines the problem as due to lack of scholarships and/or financial limitations on the part of students. The Dean’s definition would in itself have contained the “solution” that more scholarships and financial assistance be extended to masters program students, the result of which rule out other solutions for consideration.

2. The second phase in problem solving is the *gathering of information*. Once a problem is explicitly defined, one should begin to gather information about it. The information may be of many kinds. The detective may call his or her information “clues”; the doctor speak of “symptoms”; the scientist, of “data”; the layperson or government leader, of “facts.” Adequate and accurate information is essential to sound decisions. In general, the more information you have on which to base your decision, the more likely it is that the decision will be sound.

3. The third phase of decision-making is the *formation of tentative conclusions* which represent solutions to the problem. This can be done as soon as we have

enough information to suggest some possible answers. Solutions at this stage are only tentative and so we shouldn't allow ourselves to be carried away by them.

The objective in this phase is not to settle on one conclusion but rather to formulate as many plausible tentative conclusions as possible. The more tentative conclusions we produce the more likely we are to conclude a sound one. Forming several tentative conclusions is the best safeguard against the dangers of accepting or acting upon a proposed conclusion without adequate evidence. In this phase, it is desirable to give attention to every idea that comes to the mind. This is important because often, ideas you impatiently reject as wild or irrelevant turn out to be solutions of problems or important clues to solutions.

4. The fourth phase is *testing of tentative conclusions*. The objective of this phase is to “criticize” all tentative conclusions by assessing their reliability. All tentative conclusions are reached through some kind of inference, a process of reasoning by which they are derived from evidence or available facts.

5. The fifth and final phase is *Evaluation and Decision*. The objective of this phase is to determine whether you have found any workable solutions to your problem and, if so, to select the best of them. This phase involves assessing the reliability of solutions based on the testing done in phase four. When you begin testing tentative conclusions by appropriating methods, you will soon discover that completely reliable conclusions are rare. Usually there will be weaknesses either in the evidence or in the inferences or in both. In practical matters, the best we can hope

for is high degree reliability. If we delayed making a decision until we reached absolute reliability, we would dwell forever in the limbo of decision by indecision.

Problem solving skills

As earlier indicated a problem can be any task or assignment or project that need completion. Usually, it involves several different steps, phases or stages as indicated earlier. Some of the phases will require specific methods that are particular to a given subject area.

There are generic skills and processes that contribute to success for any kind of ‘problem’, whether writing an assignment, conducting a student project, undertaking projects at work, or taking on an undertaking in your personal or professional life.

Usually, to approach most problems one needs to:

- *Define the task clearly:* What exactly is required?
- *Set priorities:* What must be done first? What can be left until later?
- *Develop an appropriate strategy:* what steps must you take to address the task?

- *Use experience from similar problems:* what do you already know or what have you already done that would offer a starting place or guidance on how to approach the current problem?

- *Set targets:* what steps must you accomplish by when? How will you know you have achieved each target? How will you measure your progress?

- *Develop an action plan:* List all the steps necessary to achieve each target. Identify the best order for accomplishing each step and a deadline for each.
- *Get started:* Do not wait until the last minute, start early on the tasks that you can begin straight away. Keep yourself focused and motivated.
- *Monitor your performance against targets and indicators:* Check regularly whether you are meeting your targets and revise your action plan accordingly.
- *Evaluate your performance:* How well did you achieve your targets? What did you learn that will be of use to you for future problems and tasks?

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